



# Vojtěch Kubáč

C++ engineer • CFD, FEA, applied ML

## Experience

2024–today **Software engineer, Ententee**

- Applied-computing work alongside my role at CFD Support.
- 

2019–today **C++ software engineer, CFD Support s.r.o., Prague**

Own multiple modules of TCAE, a cross-platform CAE desktop application built around ParaView/VTK and OpenFOAM. Core stack: C++23, CMake, Qt, GitLab CI.

- **TFEA** — structural/thermal FEA module: CalculiX integration, CFD-to-FEA load mapping for FSI-style pipelines, modal analysis, ParaView-centric workflow UI.
- **TCAA** — computational aeroacoustics: FW-H / Farassat 1A far-field pipelines from CFD surfaces, SPL and octave-band processing, directivity plots, WAV export.
- **TBASE** — SQLite-backed simulation database: reproducible archive of simulations, parameters and artifacts; groundwork for surrogate modelling over parametric design spaces.
- **TMESH** — FEA volume meshing from STL/STEP via NetGen and Gmsh with quality controls and multi-component assemblies.
- Cross-module contributions to TCFD / TOPT (OpenFOAM orchestration, DoE / DIRECT / EGO-style optimisation), DevOps (GitLab CI, Linux/Windows packaging, CTest + headless Qt), and early architecture for TBRAIN, an in-app RAG assistant for TCAE.

## Selected projects

IBKR trading bot Personal Python automation around Interactive Brokers: modular workflows, operational safety, logging and failure visibility. [github.com/VojtechKubac/ibkr-trading-bot](https://github.com/VojtechKubac/ibkr-trading-bot)

This portfolio Single-page Svelte 4 / Vite / Tailwind site with EN/DE/CS i18n. [github.com/VojtechKubac/homepage](https://github.com/VojtechKubac/homepage)

Master's thesis Public code and artifacts from the master's thesis on fluid-structure interaction. [github.com/VojtechKubac/MasterThesis](https://github.com/VojtechKubac/MasterThesis)

## Education

2017–2020 **Master, Charles University, Prague, Mathematical Modeling in Physics and Technology**  
Thesis: *Comparison of possible formulations of fluid-structure interactions with application in biomechanics*, supervised by RNDr. Jaroslav Hron, Ph.D.

2018–2019 **Erasmus, Technical University Munich, Germany, Faculty of Mathematics**  
Machine learning, deep learning, high-performance computing.

2014–2017 **Bachelor, Charles University, Prague, General Mathematics**

Bratranců Veverkových 646 – 53002, Pardubice – Czech Republic

☎ (+420) 737 059 153 • ✉ [vojta.kubac@gmail.com](mailto:vojta.kubac@gmail.com)

🌐 [vojtakubac.example.com](https://vojtakubac.example.com) • **in** Vojtěch Kubáč • [VojtechKubac](https://github.com/VojtechKubac)

## Computer skills

Languages C++ (C++23), Python, Bash  
CAE & scientific OpenFOAM, CalculiX, NetGen, Gmsh, ParaView, VTK, FEniCS  
Platform Qt, CMake, SQLite, GitLab CI, Linux, Git  
AI / ML PyTorch, TensorFlow, scikit-learn; RAG / LLM integration

## Languages

Czech Native speaker  
English C1  
German B1